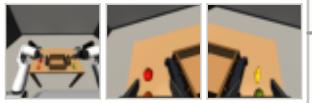
attention → distance constraint | action chunk → initial trajectory | closed loop | geometry | external module

LOCAL PC · MuJoCo

INPUT

Instruction
"Pick up the apple and place it in the basket."

RGB × 3



Robot state $q \in \mathbb{R}^{16}$
(arm 7 + gripper) × 2

RGB-D × 3
depth 480×640
K, $T_{base \leftarrow cam}$
capture-time pose

next observation

MuJoCo sim
execute 8 / 50 steps
15 Hz

websocket · SSH tunnel

GPU SERVER · serve_safe.py

π0.5 + AG3S + trajopt · one process

VLA · π0.5

16D LoRA fine-tune

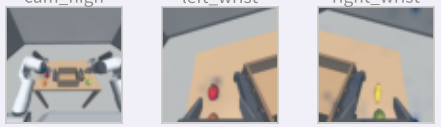
VLM prefix
PaliGemma 2B
• image tokens 3 × 256
• instruction
• state

Action expert · flow matching · 10 Euler steps

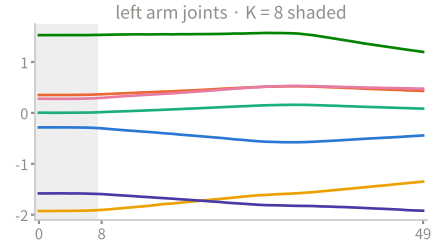
$A_{t-\delta} = A_t - \delta\, v_\theta(A_t, t), \quad \delta = 0.1$

noise → k=0 (t=1.0) → k=1 (t=0.9) → k=2 (t=0.8) → k=3 (t=0.7) → k=4 (t=0.6) → k=5 (t=0.5) → k=6 (t=0.4) → k=7 (t=0.3) → k=8 (t=0.2) → k=9 (t=0.1)

Attention readout · k=0
layer 8 · head 2
query: action token
key: image tokens



Action chunk
$A \in \mathbb{R}^{50 \times 16}$



attention 3 × 16×16

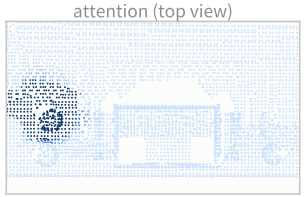
AG3S

Attention-Guided 3D Safety Scene

① Robot mask
self-filter
FK → URDF spheres
remove robot pixels

points

② Attention lifting
2D attention → 3D points



③ Target grounding
3D clusters → Target vs Obstacle



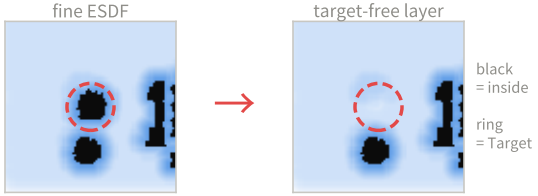
masked depth

action chunk → Q^0

Target exclusion

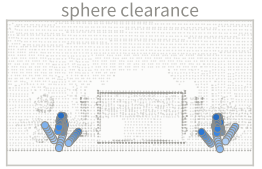
external · cuRobo

④ TSDF → ESDF
coarse 20 mm · fine 5 mm



ESDF

⑤ Distance adapter
distance · gradient
margin by label

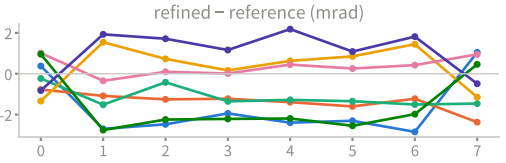


d, ∇d

query

external · trajopt

⑥ Trajectory optimization
SQP + OSQP · init $Q^0 = A$
s.t. $d - r - \text{margin} \geq 0$



OUTPUT
Q^*
+ verdict

closed loop · execute K = 8 steps @ 15 Hz → next observation